

Smart Parking Platform

Project Management Plan — Weekly Deliverables

Final Project Due: December 4, 2026

Week 1: Research

What Already Exists?

Several parking apps already exist, including ParkMobile, Flowbird, and ParkWhiz. These solutions address the problem by offering different ways to search, book, and pay for parking.

What They Do Right

The parking apps currently available to customers provide the following services:

- Search for parking
- Navigate to the selected parking via GPS
- Book a parking space
- Pay for parking online
- Receive notifications about parking

What We Will Do Better

Our Smart Parking Platform will include all the advantages of the existing solutions while providing additional services such as parking garage management.

In other words, the customers will be able to

- Find parking,
- View available parking spaces in real-time,
- Get directions to the parking,
- Book and pay for parking, and
- Receive notifications.

Additionally, the parking operators will be able to view the occupancy data and reports regarding their parking facilities.

Conclusion

Overall, the existing solutions address the problem of finding and paying for parking. Still, our task is to offer a comprehensive and easy-to-use platform that will suit both drivers searching for parking and the owners of the parking garage.

Vision and Scope

How the Project Was Acquired

WE ARE \$oftware ¢orp. was approached by parking facility operators who were interested in improving their operations. They needed a solution that would help them provide better customer service by enabling motorists to find parking spaces, settle payments conveniently, and offer them a clear insight into the workings of the parking system.

Company Introduction

WE ARE \$oftware ¢orp. is an imaginary software design organization with an unlimited budget. The company specializes in designing software solutions that make people's lives easier. In this case, our experts are going to create an application for the client that will improve customer experience and give the owners insight into their business data.

Project Overview

The acquired project is described as the development of a Smart Parking Platform that will be used by drivers to find parking, settle payments, reserve spaces, and access related services. This platform will be distributed as web and mobile applications where motorists can check parking availability, get directions, pay fees, and make reservations. The applications will feature parking search, navigation, reservations, and digital payment systems, and provide drivers with convenient and reliable parking experience.

The software will also offer back-end tools and administrative dashboards for parking operators and give them greater control and efficiency in managing their operations. The given project aims to create a smart parking solution for drivers and admins that will offer a seamless and convenient experience while providing motorists with the ability to find the best option available.

SRS Draft

Purpose

The proposed Smart Parking Platform will offer an innovative solution for both drivers, looking for a place to park their cars conveniently, and parking owners, managing their storage services efficiently.

Functional Requirements

The software will be required to provide the following features:

- User registration and authorization;
- Search for available parking;
- Check the current state of parking;
- See the parking details on the map;
- Check the prices;
- Make a reservation;
- Make a payment;
- See the history of reservations and payments;
- Get notifications about the reservation status change or parking details;
- Get directions to the car park.
- Cancel the reservation if it is possible;
- Report incorrect information about parking.

The system will also allow parking owners to manage parking spaces, check their current state, review reports, and update parking details and prices.

Week 2 Deliverable 1: Work Breakdown Structure

This week focuses on creating the **Work Breakdown Structure (WBS)** for the Smart Parking Platform and drafting the **project timeline, including the Gantt chart**.

WBS	Work Package
1.0	Smart Parking Platform
1.1	Research & Project Initiation
1.1.1	Research existing parking solutions
1.1.2	Define project vision & scope
1.1.3	Develop initial SRS
1.2	Requirements Analysis
1.2.1	Driver requirements
1.2.2	Parking operator requirements

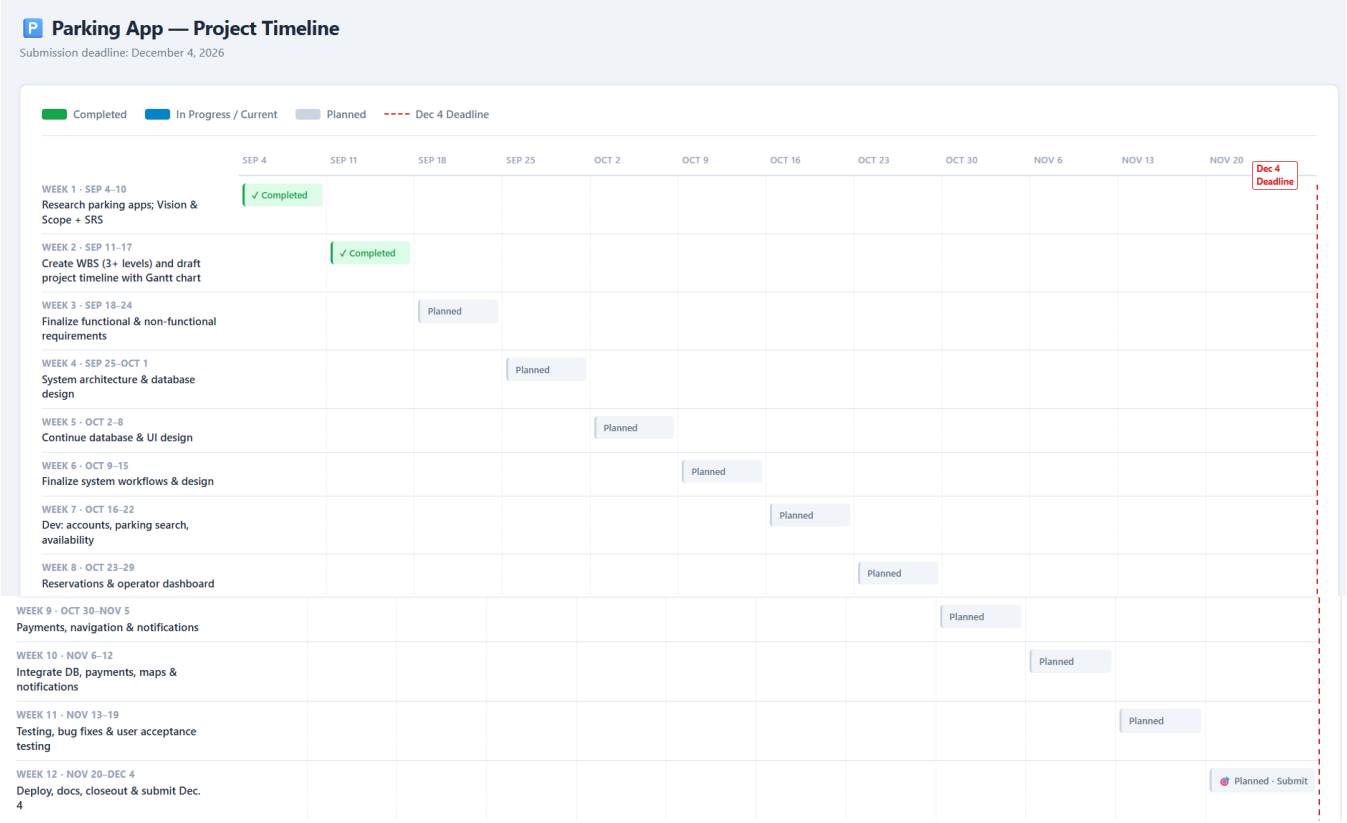
1.2.3	Functional and non-functional requirements
1.3	System Design
1.3.1	Design system architecture
1.3.2	Design database
1.3.3	Design user interface
1.3.4	Design system workflows
1.4	Application Development
1.4.1	User accounts and authentication
1.4.2	Parking search and availability
1.4.3	Reservation system
1.4.4	Payment system
1.4.5	Navigation and notifications
1.4.6	Operator dashboard
1.5	System Integration
1.5.1	Database integration
1.5.2	Payment integration
1.5.3	GPS/map integration
1.5.4	Notification integration
1.6	Testing & Quality Assurance
1.6.1	Functional and system testing
1.6.2	Security and performance testing
1.6.3	User acceptance testing
1.6.4	Fix defects and retest
1.7	Deployment & Implementation
1.7.1	Prepare production environment
1.7.2	Deploy web/mobile applications
1.7.3	Training and documentation
1.7.4	Go-live
1.8	Project Closeout
1.8.1	Final stakeholder acceptance
1.8.2	Lessons learned
1.8.3	Project handoff and closeout

Week 2 Deliverable 2: Project Timeline by Week

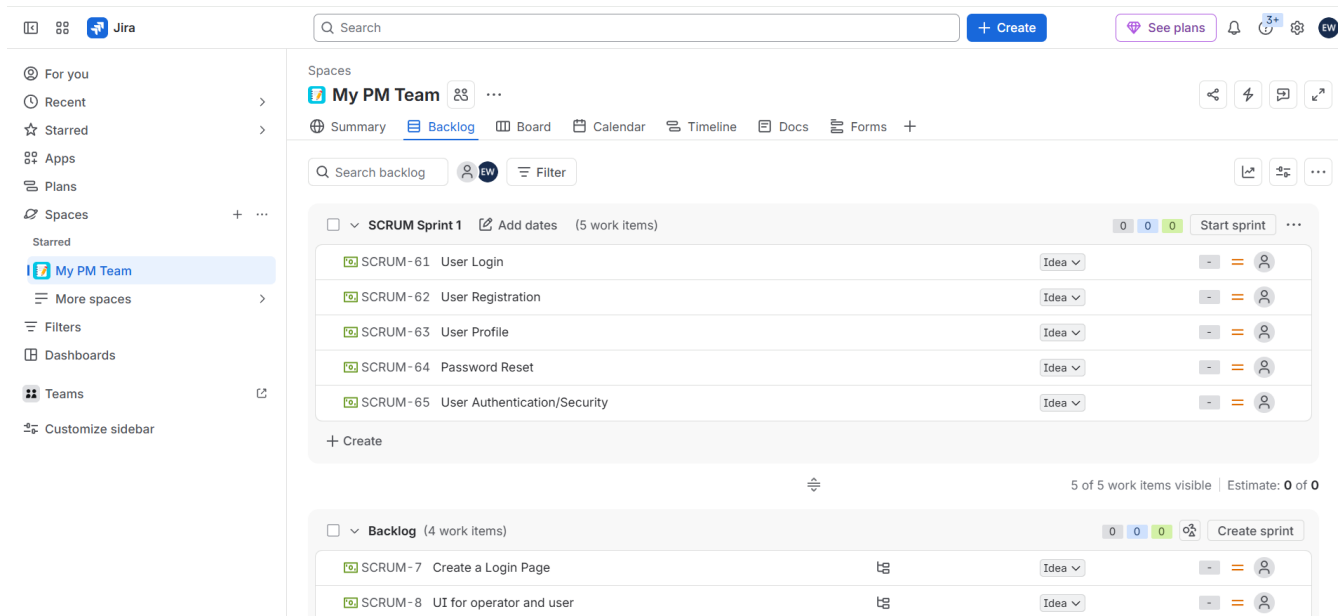
Week	Planned Work	Status
Week 1	Research existing parking apps; complete Vision & Scope and initial SRS.	Completed / current
Week 2	Create WBS (3+ levels) and draft project timeline with Gantt chart	Completed
Week 3	Finalize and organize functional/non-functional requirements.	Planned
Week 4	Start system architecture and database design.	Planned
Week 5	Continue database and user-interface design.	Planned
Week 6	Finalize system workflows and design.	Planned
Week 7	Begin development: accounts, parking search, and availability.	Planned
Week 8	Continue reservations and operator dashboard.	Planned
Week 9	Complete payments, navigation, and notifications.	Planned
Week 10	Integrate database, payment, maps, and notifications.	Planned
Week 11	Test the system, fix issues, and complete user acceptance testing.	Planned

Week 12	Deploy, finalize documentation, complete closeout, and submit by Dec. 4.	Planned
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Gantt Chart



Week 3: Create Backlog (Jira)



Week 4: Risk Management & Communication Plan

Technical Risks:

1. **Parking availability errors:** The system may display a space as available even when it is occupied.
2. **GPS problems:** The GPS may provide incorrect locations or directions.
3. **Payment failures:** Users may experience errors while making payments through the app.
4. **System bugs:** The development team may encounter bugs that do not have a proper fix.

Financial Risks:

1. **Higher API costs:** GPS, payment, or other API tools may become expensive.
2. **Cloud service costs:** Some cloud services, including Hosting and storing data, may be costly.
3. **Additional cost:** The project may incur some additional costs that were not budgeted.
4. **Maintenance costs:** There may be some maintenance costs associated with the system.

People's Risks:

1. **Poor Communication:** There may be poor information sharing among the team members.
2. **Uneven workload:** Some team members may have more work than others.
3. **Absence:** A team member may be absent and leave their work unfinished.

4. **Poorly defined responsibilities:** The responsibilities of some team members may not be clearly defined.

Schedule Risks:

1. **Log tasks:** Some tasks may take longer than anticipated.
2. **Testing Errors:** The discovery process of errors may cause project delays.
3. **Changed requirements:** Some requirements may change and add up extra work.
4. **Rework:** Some work may have to be reworked due to errors.

Risk Register:

ID	Risk	Category	Probability	Impact	Description
R1	Parking availability error	Technical	Medium	High	Test the availability updates regularly and compare them with actual parking details. In case of malfunction, update the system.
R2	GPS malfunctions	Technical	Medium	High	Test GPS and navigation regularly in different geographical locations.
R3	Payment errors	Technical	Medium	High	Check, test, and verify the payment procedures before launch and create an alternate process if payment fails.
R4	Bugs in the system	Technical	High	Medium	Test every function, identify, and remove bugs during regular tests or while using the software.
R5	Increased API costs	Financial	Medium	Medium	Evaluate and analyze the cost of using the API and track the expenses during the project.
R6	Increased Cloud costs	Financial	Medium	Medium	Keep track of the cloud costs and usage.
R7	Unexpected development cost	Financial	Medium	High	Keep track of the project budget and approve all unexpected expenses.
R8	Maintenance and support cost	Financial	Medium	Medium	The project plan should have provisions for future maintenance, updates, and technical support.
R9	Ineffective communication with team members	People	Medium	High	Conduct regular meetings and always update the Jira system with project details.
R10	Ineffective allocation of workloads	People	Medium	Medium	Re-evaluate the work distribution and reallocate tasks as necessary.
R11	Absence of experienced employees	People	Medium	Medium	Update all team members on project details so that any available member can take up the responsibilities of an absent employee.
R12	Delayed tasks	Schedule	High	Medium	Monitor and control the schedule to ensure that all tasks are completed within the deadline or reallocated to meet the deadline.
R13	Delayed testing	Schedule	Medium	High	Test the software at an early stage and resolve all major issues as they are identified.
R14	Changed requirements	Schedule	Medium	High	Go through the project requirements changes with the team before implementation.

Communication Plan:

Team/Group	Frequency	Purpose	Reporting Method
Project Team	Weekly	To review the project's progress, tasks, and potential risks.	Jira + Meeting notes
Project Manager	Twice a week	To monitor the deadlines, risks, and individual responsibilities.	Jira + Risk register
Technical Team	Weekly	To address technical challenges, bugs, GPS, payments, and parking availability.	Jira + Team chat
Financial Team	Every 2 weeks	To assess the expenses related to API, cloud, development, and maintenance.	Budget report
Team Leads	Weekly	To evaluate the team's workload, communication effectiveness, and availability.	Jira + Team chat
Testing Team	Weekly	To track testing progress, identify bugs, and discuss delays.	Jira + Testing reports
Design/UI Team	Weekly	Assess design progress, prototype changes, and potential rework areas.	Jira + Screenshots
Stakeholders	Every 2 weeks	To report on the project's progress, risks, and requirement changes.	Progress report + Presentation
Entire Team	As needed	To deliberate urgent issues or project-related changes.	Team chat + Jira